



Republic of the Philippines
Department of Agriculture
BUREAU OF FISHERIES AND AQUATIC RESOURCES
REGIONAL FISHERIES LABORATORY XII
J. Catolico St., Lagao, General Santos City

Document Type Work Instruction	Revision No: 0	Date Adopted: 13 Aug 2019	Page No: Page 1 of 9
Document Code: WI-01-MIC-06	COLIFORMS AND E. COLI IN FISH AND FISHERY PRODUCTS		

1. OBJECTIVE

To determine the presence of coliform bacteria and *E. coli* in fish and fishery products using the 3-tube Most Probable Number (MPN) method.

2. SCOPE

This instruction is applicable in the determination of coliform bacteria and *E. coli* organism in fish and fishery products.

3. RESPONSIBILITY

Laboratory Analyst.

4. APPARATUS/EQUIPMENT/CONSUMABLES

- 4.1 Alcohol lamps / Fireboy
- 4.2 Disinfectant (70% solution isopropyl alcohol)
- 4.3 Autoclave
- 4.4 Autoclave tape
- 4.5 Top loading Balance
- 4.6 Aluminum Foil
- 4.7 Tissue paper or paper towel
- 4.8 Spatula
- 4.9 Forceps – stainless steel
- 4.10 Scissors – stainless steel
- 4.11 Graduated cylinder (100 ml and 500 ml, Pyrex)
- 4.12 Media bottles (500 ml and 1000 ml, Pyrex)
- 4.13 Dilution bottles with screw caps (160 ml, Pyrex)
- 4.14 Sterile Pipets (10 ml)
- 4.15 Sterile Pipettor

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Document Code: WI-01-MIC-06	COLIFORMS AND E. COLI IN FISH AND FISHERY PRODUCTS		

- 4.16 Sterile Pipet tips (100 – 1000 µl and 10 – 200 µl)
- 4.17 Sterile Petri dishes with cover (100x15mm, Pyrex)
- 4.18 Test Tubes with autoclavable caps (10 ml, 15 ml, and 20 ml, Pyrex)
- 4.19 Durham tubes (6x50 mm, Pyrex)
- 4.20 Test tube racks
- 4.21 Stomacher
- 4.22 Stomacher bags
- 4.23 Incubators maintained at 35°C and (45.5 shall be deleted) ± 0.2°C
- 4.24 Covered water bath, with circulating system to maintain temperature of 45.5 ± 0.2°C.
- 4.25 Inoculating loops and needles
- 4.26 Biosafety Cabinet

5. CULTURE MEDIA AND REAGENTS

5.1 All information regarding media preparation is placed in LF-W04-01, Reagent Preparation Logbook.

5.2 Butterfield's Phosphate-Buffered Dilution Water

Suspend 34 g of KH₂PO₄ to 500 milliliter of distilled water. Adjust pH to 7.2 with 1 N NaOH. Bring volume to 1 liter with distilled water. Sterilize 15 min at 121°C. Store in a refrigerator at 4°C.

Dilution Blanks

Take 1.25 ml of above stock solution and bring volume to 1 liter with distilled water. Dispense into bottles to 90 or 99 ± 1 ml. Sterilize 15 min at 121°C.

5.3 Pre-enrichment and Selective Media

5.3.1 Lauryl Tryptose Broth (LST)

Suspend 35.6 g in 1 liter of demineralized water. Dispense 10 ml volumes in 20 ml test tubes with inverted Durham tubes. Cover the tubes with autoclavable caps. Autoclave 15 min at 121°C. Final pH, 6.8 ± 0.2.

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Document Code: WI-01-MIC-06	COLIFORMS AND E. COLI IN FISH AND FISHERY PRODUCTS		

5.3.2 Brilliant-Green Lactose Bile (BGLB) 2% Broth

Suspend 40 g in 1 liter of demineralized water. Dispense 10 ml volumes in 20 ml test tubes with inverted Durham tubes. Cover the tubes with autoclavable caps. Autoclave 15 min at 121°C. Final pH, 7.2 ± 0.1.

5.3.3 EC Medium

Suspend 37 g in a liter of demineralized water. Dispense 8 ml volumes in 15 ml test tubes with inverted Durham tubes. Autoclave 15 min at 121°C. Final pH, 6.9 ± 0.2.

5.4 Differential Medium

5.4.1 Levine's Eosin Methylene Blue (L-EMB) Agar

Suspend 36 g of powder in a liter of purified water. Mix thoroughly and dissolve completely by heating in a boiling water bath. Autoclave 15 min at 121°C. Final pH, 7.1 ± 0.2.

5.5 Media for Biochemical Tests

5.5.1 Tryptone (tryptophane) Broth, 1%

Suspend 10 g in a liter of distilled water. Dissolve completely. Dispense 5 ml volumes in 10 ml test tubes. Autoclave 15 min at 121°C. Final pH, 6.9 ± 0.2.

5.5.2 MR-VP Medium

Dissolve 17 g in 1 liter of demineralized water. Dispense 5 ml volumes in 10 ml test tubes. Autoclave 15 min at 121°C. Final pH, 6.9 ± 0.2.

5.5.3 Simmons Citrate Agar (SCA)

Suspend 24.28 g in a liter of demineralized water. Heat gently with occasional agitation. Boil for 1-2 minutes until agar dissolves. Fill 10 ml or 20 ml screw-cap tubes 1/3 full. Autoclave at 121°C for 15 minutes. Before medium solidifies, incline tubes to obtain 4-5 cm slants and 2-3 cm butts. Final pH, 6.8 ± 0.2.

5.5.4 Plate Count Agar (PCA) Slants

Suspend 23.5 g of the medium in one liter of distilled water. Mix thoroughly. Heat with frequent agitation and boil for 1 minute to completely dissolve the powder. Fill 10 ml or 20 ml screw-cap tubes 1/3 full. Sterilize 15 min at 121

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Document Type Work Instruction	Revision No: 0	Date Adopted: 13 Aug 2019	Page No: Page 4 of 9
Document Code: WI-01-MIC-06	COLIFORMS AND E. COLI IN FISH AND FISHERY PRODUCTS		

°C. Before medium solidifies, incline tubes to obtain 4-5 cm slants and 2-3 cm butts. Final pH, 7.0 ± 0.2 .

5.6 Reagents

5.6.1 Erlich's Reagent

Dissolve 2.0 g of p-dimethylaminobenzaldehyde in 190 ml absolute ethyl alcohol. Add 40 ml concentrated HCl.

5.6.2 Kovac's Reagent

Dissolve 10 g of p-dimethylaminobenzaldehyde in 150 ml pure amyl or isoamyl alcohol and slowly add 50 ml concentrated HCl.

5.6.3 Methyl Red Indicator

Dissolve 0.1 g methyl red in 300 ml of 95% ethyl alcohol and dilute to 500 ml with distilled water.

5.6.4 α -naphthol solution

Dissolve 5 g purified α -naphthol in 100 ml absolute ethanol.

5.6.5 Potassium hydroxide (KOH) solution

Dissolve 40 g potassium hydroxide in 100 ml distilled water.

6. PROCEDURE

6.1 PRESUMPTIVE TEST FOR COLIFORMS, FECAL COLIFORMS AND *E. COLI*

6.1.1 Weigh 25 g sample into a sterile stomacher bag. Add 225 ml sterile Butterfield's phosphate-buffered dilution water to make a 10^{-1} dilution and homogenize using the stomacher for 2 minutes at 230 rpm.

6.1.2 Prepare decimal dilutions of food homogenate, 10^{-2} and 10^{-3} respectively by sequentially transferring 10 ml suspension into 90 ml diluent. Shake all dilutions 25 times in 30 cm arc or vortex mix for 7 s to ensure uniform distribution of microorganisms.

6.1.3 For each dilution transfer 1 ml of aliquots into each marked 3 LST tubes with inverted Durham tubes.

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Document Code: WI-01-MIC-06	COLIFORMS AND E. COLI IN FISH AND FISHERY PRODUCTS		

6.1.4 Incubate LST tubes at 35°C for 24 ± 2 h. Examine tubes and record reactions such as displacement of medium in Durham tubes or effervescence when tubes are gently agitated.

6.1.5 Re-incubate gas-negative tubes for an additional of 24 h. Examine and record reactions again at 48 ± 2 h.

6.1.6 Perform confirmed test on all presumptive positive (gassing) tubes.

6.2 CONFIRMED TEST FOR COLIFORMS

6.2.1 Transfer a loopful of suspension from each positive LST tube into a tube of BGLB broth.

6.2.2 Invert tubes to ensure Durham tubes does not contain gas bubbles.

6.2.3 Incubate the BGLB broth tubes at 35°C for 48 ± 2 h.

6.2.4 Examine for gas production.

6.2.5 Read Most Probable Number of coliforms based on proportion of confirmed gassing LST tubes for 3 consecutive dilutions from MPN Table 2. Record results.

6.3 CONFIRMED TEST FOR FECAL COLIFORMS AND *E. COLI*

6.3.1 Transfer a loopful of suspension from each gassing LST tube to a tube of EC broth.

6.3.2 Incubate EC tubes at 45.5 ± 0.2 °C for 24 ± 2 h.

6.3.3 Examine for gas production at 24 h and if negative, examine again at 48 ± 2 h.

6.3.4 Any tube showing gas production is considered positive for the presence of presumptive *E. coli*.

6.3.5 Read Most Probable Number of fecal coliforms based on proportion of confirmed gassing EC tubes for 3 consecutive dilutions from MPN Table 2. Record results.

6.4 COMPLETED TEST FOR *E. COLI*

6.4.1 Gently agitate each gassing EC tube and streak for isolation a loopful to L-EMB agar plate.

6.4.2 Incubate at 35°C for 18 - 24 h.

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Document Type Work Instruction	Revision No: 0	Date Adopted: 13 Aug 2019	Page No: Page 6 of 9
Document Code: WI-01-MIC-06	COLIFORMS AND E. COLI IN FISH AND FISHERY PRODUCTS		

6.4.3 Examine the plates for suspicious *E. coli* colonies, i.e. black or dark centered with or without the greenish metallic sheen.

6.4.4 Transfer up to 5 suspicious colonies from each L-EMB plates to PCA slants.

6.4.5 Incubate for 18 – 24 h at 35°C and use for further testing.

Note: Identification of any 1 of the 5 colonies as *E. coli* is sufficient to regard that EC tube as positive; hence, not all 5 isolates may need to be tested.

6.4.6 Perform Gram stain. All cultures appearing as Gram-negative, short rods should be tested for the IMViC reactions and also re-inoculated back into LST to confirm gas production.

6.5 IMViC TESTS

6.5.1 Indole Production

Inoculate tube of Tryptone Broth and incubate at 35°C for 24 ± 2 h. Test for indole by adding 1 ml xylene, shake well and allow xylene to rise. Add 0.5 ml Erlich's reagent. Alternatively add 0.5 ml of Kovac's reagent directly to the TB culture. Appearance of distinct red color in upper layer indicates positive indole test.

6.5.2 Voges-Proskauer (VP)

Inoculate tube of MRVP broth and incubate for 48 ± 2 h at 35°C. Transfer 1 ml to 13 x 100 mm tube. Add 0.6 ml α-naphthol solution and 0.2 ml 40% KOH and shake. Add a few crystals of creatine. Shake and let stand for 2 h. Test if positive if eosin pink color develops.

6.5.3 Methyl Red

After VP Test, incubate MRVP tube additional 48 ± 2 h at 35°C. Add 5 drops of methyl red solution to each tube. Distinct red color is a positive test. Yellow is a negative reaction.

6.5.4 Citrate Test

Streak and stab tube of Simmons citrate agar slant with the isolate. Incubate for 96 h at 35°C. Growth on the medium with blue coloration is a positive reaction for citrate utilization.

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Document Type Work Instruction	Revision No: 0	Date Adopted: 13 Aug 2019	Page No: Page 7 of 9
Document Code: WI-01-MIC-06	COLIFORMS AND E. COLI IN FISH AND FISHERY PRODUCTS		

6.5.5 Interpret the results of the IMViC tests according to the reactions of coliforms and other related microorganisms in Table 1.

Table 1. Reactions of *E. coli* and other related microorganisms in IMViC test.

Test	<i>Escherichia coli</i>	<i>Citrobacter freundii</i>	<i>Escherichia hermannii</i>	<i>Escherichia vulneris</i>	<i>Enterobacter aerogenes</i>	<i>Klebsiella</i>	<i>Pseudomonas</i>	<i>Aeromonas</i>
Indole	+	-	+	-	-	±	-	+
Methyl Red	+	+	±	+	-	-	-	-
Voges-Proskauer	-	-	-	-	+	-	-	+
Citrate	-	+	-	-	+	+	+	+

6.6 Quantitative Determination of *E. coli*

All cultures that give IMViC patterns of +++- are considered to be *E. coli*. Read Most Probable Number based on proportion of EC tubes in 3 successive dilutions that contain *E. coli* from Table 2.

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Document Type Work Instruction	Revision No: 0	Date Adopted: 13 Aug 2019	Page No: Page 8 of 9
Document Code: WI-01-MIC-06	COLIFORMS AND E. COLI IN FISH AND FISHERY PRODUCTS		

Table 2. For 3 tubes each at 0.1, 0.01, and 0.001 g inocula, the MPNs per gram and 95 percent confidence intervals.											
Pos. tubes			MPN/g	Conf. lim.		Pos. tubes			MPN/g	Conf. lim.	
0.10	0.01	0.001		Low	High	0.10	0.01	0.001		Low	High
0	0	0	<3.0	--	9.5	2	2	0	21	4.5	42
0	0	1	3.0	0.15	9.6	2	2	1	28	8.7	94
0	1	0	3.0	0.15	11	2	2	2	35	8.7	94
0	1	1	6.1	1.2	18	2	3	0	29	8.7	94
0	2	0	6.2	1.2	18	2	3	1	36	8.7	94
0	3	0	9.4	3.6	38	3	0	0	23	4.6	94
1	0	0	3.6	0.17	18	3	0	1	38	8.7	110
1	0	1	7.2	1.3	18	3	0	2	64	17	180
1	0	2	11	3.6	38	3	1	0	43	9	180
1	1	0	7.4	1.3	20	3	1	1	75	17	200
1	1	1	11	3.6	38	3	1	2	120	37	420
1	2	0	11	3.6	42	3	1	3	160	40	420
1	2	1	15	4.5	42	3	2	0	93	18	420
1	3	0	16	4.5	42	3	2	1	150	37	420
2	0	0	9.2	1.4	38	3	2	2	210	40	430
2	0	1	14	3.6	42	3	2	3	290	90	1,000
2	0	2	20	4.5	42	3	3	0	240	42	1,000
2	1	0	15	3.7	42	3	3	1	460	90	2,000
2	1	1	20	4.5	42	3	3	2	1100	180	4,100
2	1	2	27	8.7	94	3	3	3	>1100	420	--

7. REFERENCE

Bacteriological Analytical Manual, Edition 8, Revision A, 1998. Chapter 4. Authors: Peter Feng, Stephen D. Weagant (ret.), Michael A. Grant (dec.), William Burkhardt

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Document Code: WI-01-MIC-06	COLIFORMS AND E. COLI IN FISH AND FISHERY PRODUCTS		

8. FORMS

LF W04-01, Reagent and Media Preparation Logbook

LF W04-MIC-03, Analyst Worksheet for Bacteriological Examination of Fish and Fishery Products

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